

- Estimare parametrilor necunoscuți (coeficienți) și zgomot alb) ARX(1,1) sau MA(1) ARX(1,1) $\Rightarrow A(z^{-1})y[k] = B(z^{-1})u[k] + e[k]$

Curs 7 / Exercițiu 6.1

$$y[k] = \varphi^T[k] \theta + e[k]$$

↳ formă de rețea liniară

$$\begin{bmatrix} \varphi^T[k] \\ \theta^T \end{bmatrix} = \begin{bmatrix} -y[k-1] & u[k-1] \\ a & b \end{bmatrix}$$

• Estimare CMMP:

$$\hat{\theta}_N = \left[\frac{1}{N} \sum_{k=1}^N \varphi[k] \varphi^T[k] \right]^{-1} \left[\frac{1}{N} \sum_{k=1}^N \varphi[k] y[k] \right]$$

$$NR_N = \sum_{k=1}^N \begin{bmatrix} -y[k-1] \\ u[k-1] \end{bmatrix} \begin{bmatrix} -y[k-1] & u[k-1] \end{bmatrix} = \begin{bmatrix} -\sum_{k=1}^N y^2[k-1] & -\sum_{k=1}^N y[k-1]u[k-1] \\ -\sum_{k=1}^N u[k-1]y[k-1] & \sum_{k=1}^N u^2[k-1] \end{bmatrix}$$

$$\pi_{y,u} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{N-1} \sum_{k=2}^N y[k]u[k-1] \quad NR_N^T = \begin{bmatrix} -\sum_{k=2}^N y[k]y[k-1] & \sum_{k=2}^N y[k]u[k-1] \end{bmatrix}$$

= produs (IC) + faptul că fiecare termen are N-1 termeni

$$\frac{N}{N-1} R_N = \begin{bmatrix} \pi_{yy}^{N-1}[\omega] & -\pi_{uy}^{N-1}[\omega] \\ -\pi_{uy}^{N-1}[\omega] & \pi_{uu}^{N-1}[\omega] \end{bmatrix}; \quad \frac{N}{N-1} \pi_N = \begin{bmatrix} -\pi_{yy}^N[\omega] \\ \pi_{y,u}^N[\omega] \end{bmatrix}$$

$$\left[\frac{1}{N-1} \sum_{k=2}^N y^2[k-1] = \frac{1}{N-1} \sum_{k=1}^{N-1} y^2[k] = \pi_{yy}^{N-1}[\omega] \right]$$

$$\Rightarrow \hat{\theta}_N = R_N^{-1} \pi_N = \left(\frac{N}{N-1} R_N \right)^{-1} \cdot \left(\frac{N}{N-1} \pi_N \right)$$

Result:

$$\hat{\theta}_N = \frac{1}{\Delta_N} \begin{bmatrix} \pi_{\mu}^{N-1} [0] & \pi_{y\mu}^{N-1} [0] \\ \pi_{y\mu}^{N-1} [0] & \pi_{yy}^{N-1} [0] \end{bmatrix} \begin{bmatrix} -\pi_y^N [1] \\ \pi_{y\mu}^N [1] \end{bmatrix}$$

$$\Delta_N = \pi_{\mu}^{N-1} [0] \pi_{yy}^{N-1} [0] - (\pi_{y\mu}^{N-1} [0])^2$$

$$\begin{cases} \hat{\theta}_N = \frac{1}{\Delta_N} \begin{bmatrix} \pi_{y\mu}^{N-1} [0] \pi_{y\mu}^N [1] - \pi_{\mu}^{N-1} [0] \pi_{yy}^N [0] \\ \pi_{yy}^{N-1} [0] \pi_{y\mu}^N [1] - \pi_y^N [1] \pi_{y\mu}^{N-1} [0] \end{bmatrix} = \begin{bmatrix} \hat{a}_N \\ \hat{b}_N \end{bmatrix} \\ \hat{J}_N^L = \frac{1}{N} \sum_{\mu=1}^N |y[\mu] - e^T[\mu] \hat{\theta}_N|^2 \end{cases}$$

Limit results

$$\lim_{N \rightarrow \infty} \hat{a}_N = \frac{\pi_{y\mu}[0] \pi_{y\mu}[1] - \pi_{\mu}[0] \pi_{yy}[1]}{\pi_{\mu}[0] \pi_{yy}[0] - (\pi_{y\mu}[0])^2}$$

$$\lim_{N \rightarrow \infty} \hat{b}_N = \frac{\pi_{yy}[0] \pi_{y\mu}[1] - \pi_{yy}[1] \pi_{y\mu}[0]}{\pi_{\mu}[0] \pi_{yy}[0] - (\pi_{y\mu}[0])^2}$$

Re write

$$\theta = \begin{bmatrix} a \\ b \end{bmatrix} = [e^T e_{\mu}] e^T [\mu]^T [e^T e_{\mu}] y [\mu]^T - [e^T e_{\mu}] e^T [\mu]^T \begin{bmatrix} e^T e_{\mu} y [\mu]^T \\ [e^T e_{\mu}] e_{\mu} [\mu]^T \end{bmatrix}^{-1}$$

$$[e^T e_{\mu}] e^T [\mu]^T = \begin{bmatrix} \pi_{yy}[0] - \pi_{y\mu}[0] \\ -\pi_{y\mu}[0] \pi_{\mu}[0] \end{bmatrix}$$

$$[e^T e_{\mu}] y [\mu]^T = \begin{bmatrix} -\pi_{yy}[1] \\ \pi_{y\mu}[1] \end{bmatrix}$$

$y[\mu] (e_{\mu}^T [\mu])$ → manual work.

$$[e^T e_{\mu}] e_{\mu} [\mu]^T = \begin{bmatrix} -\pi_{ye}[1] \\ \pi_{\mu e}[-1] \end{bmatrix} = \begin{bmatrix} 0 \\ \pi_{\mu e}[1] \end{bmatrix}$$

$$\Rightarrow \text{bei } a = \frac{\pi_{yu}[0]\pi_{ye}[1] - \pi_{u}[0]\pi_{y}[1] + \pi_{ye}[0]\pi_{eu}[1]}{\pi_{u}[0]\pi_{y}[0] - (\pi_{ye}[0])^2}$$

$$b = \frac{\pi_{y}[0]\pi_{ye}[1] - \pi_{y}[1]\pi_{ye}[0] + \pi_{y}[0]\pi_{eu}[1]}{\pi_{u}[0]\pi_{y}[0] - (\pi_{ye}[0])^2}$$

Bedingung der Konsistenz $\Leftrightarrow \theta_N = \theta$

a) $\in \{e_{it}\} \in \mathbb{R}^T$ = vektorisiert

b) $\in \{e_{it}\} \cdot e_{it-1} = 0$

c) $e = \text{normal als } (0, \sigma^2)$

• ARX(2,2) - rekursiv

$$y_{it} = \theta^T \Gamma_{it} + e_{it}$$

$$\Gamma_{it} = [-y_{it-1} \quad -y_{it-2} \quad ; \quad u_{it-1} \quad u_{it-2}]$$

$$\theta^T = [a_1 \quad a_2 \quad ; \quad b_1 \quad b_2]$$

$$\hat{\theta}_N = \begin{bmatrix} \pi_{yy}^{N-1}[0] & \pi_{yy}^N[1] & -\pi_{ye}^{N-1}[0] & -\pi_{ye}^N[1] \\ \pi_{yy}^N[1] & \pi_{yy}^{N-1}[0] & -\pi_{ye}^N[1] & -\pi_{ye}^{N-1}[0] \\ \hline -\pi_{yu}^{N-1}[0] & -\pi_{yu}^N[1] & \pi_{uu}^{N-1}[0] & \pi_{uu}^N[1] \\ -\pi_{ye}^N[1] & -\pi_{ye}^{N-1}[0] & \pi_{uu}^N[1] & \pi_{uu}^{N-1}[0] \end{bmatrix} \begin{bmatrix} \pi_{yy}^N[1] \\ -\pi_{yy}^N[0] \\ \pi_{yu}^N[1] \\ \pi_{ye}^N[0] \end{bmatrix}$$

$$\left(\left[\frac{1}{N} \sum_{i=1}^N \Gamma_{it} \Gamma_{it}^T \right]^{-1} \right)$$

$$= \left(\frac{1}{N} \sum_{i=1}^N \Gamma_{it} \Gamma_{it}^T \right)$$

$A \in \mathbb{R}^{[2;2]}$ - deforant

$$A(z^{-1}) = 1 + a_1 z^{-1} + a_2 z^{-2}$$

$$B(z^{-1}) = b_1 z^{-1} + b_2 z^{-2}$$

$$\hookrightarrow y[n] + a_1 y[n-1] + a_2 y[n-2] = b_1 u[n-1] + b_2 u[n-2] + e[n]$$

$$\hookrightarrow y[n] = -a_1 y[n-1] - a_2 y[n-2] + b_1 u[n-1] + b_2 u[n-2] + e[n]$$

$$y[n] = e^T \theta + e[n]$$

$$= [-y[n-1] \quad -y[n-2] \quad u[n-1] \quad u[n-2]] \begin{bmatrix} a_1 \\ a_2 \\ b_1 \\ b_2 \end{bmatrix} + e[n]$$

Estimate OMP:

$$y[n] = e^T \theta + e[n]$$

$$e[n] y[n] = e e^T \theta + e e[n]$$

$$\Rightarrow E\{e[n] y[n]\} = E\{e[n] e^T\} \theta + \underbrace{E\{e[n] e[n]\}}_{=0}$$

$$\theta = \left[E\{e[n] e^T\} \right]^{-1} \cdot \underbrace{E\{e[n] y[n]\}}_{q_N}$$

$$R_N^{-1} = \left(\frac{1}{N} \sum_{n=1}^N e[n] e^T[n] \right)^{-1}$$

$$\Rightarrow N R_N = \sum \begin{bmatrix} -y[n-1] \\ -y[n-2] \\ u[n-1] \\ u[n-2] \end{bmatrix} \cdot \begin{bmatrix} -y[n-1] & -y[n-2] & u[n-1] & u[n-2] \end{bmatrix}$$

$$NH_N = \sum_{n=1}^N \begin{pmatrix} y^2[n-1] & y[n-1]y[n-2] & -y[n-1]u[n-1] & -y[n-1]u[n-2] \\ y[n-2]y[n-1] & y^2[n-2] & -y[n-2]u[n-1] & -y[n-2]u[n-2] \\ -u[n-1]y[n-1] & -u[n-1]y[n-2] & u^2[n-1] & u[n-1]u[n-2] \\ -u[n-1]y[n-1] & -u[n-2]y[n-2] & u[n-2]u[n-1] & u^2[n-2] \end{pmatrix}$$

$$\frac{N}{N-2} R_N = \begin{pmatrix} \overset{N-1}{\pi_y[0]} & \overset{N-1}{\pi_y[1]} & \overset{N-1}{-\pi_{yu}[0]} & \overset{N}{-\pi_{yu}[1]} \\ \overset{N}{\pi_y[1]} & \overset{N-1}{\pi_y[0]} & \overset{N}{-\pi_{yu}[1]} & \overset{N-1}{-\pi_{yu}[0]} \\ -\overset{N-1}{\pi_{yu}[0]} & -\overset{N}{\pi_{yu}[1]} & \overset{N-1}{\pi_{uu}[0]} & \overset{N}{\pi_{uu}[1]} \\ -\overset{N}{\pi_{yu}[1]} & -\overset{N-1}{\pi_{yu}[0]} & \overset{N}{\pi_{uu}[1]} & \overset{N-1}{\pi_{uu}[0]} \end{pmatrix} \quad \begin{matrix} \text{zero padding?} \\ \text{N} \end{matrix}$$

$$NH_N = \sum_{n=1}^N e[n] \cdot y[n] = \begin{bmatrix} -\sum_{n=3}^N y[n]y[n-1] \\ -\sum_{n=3}^N y[n]y[n-2] \\ \sum_{n=3}^N y[n]u[n-1] \\ \sum_{n=3}^N y[n]u[n-2] \end{bmatrix} \begin{matrix} \frac{N-2}{N-1} \\ \frac{N-1}{N} \\ \frac{N}{N} \\ \frac{N}{N} \end{matrix} = \begin{bmatrix} -\pi_y[1] \\ -\pi_y[2] \\ \pi_{yu}[1] \\ \pi_{yu}[2] \end{bmatrix}$$

~~linear & quadratic~~
~~ca $\pi_N = \frac{N-2}{N-1}$~~
 $\Rightarrow \frac{N-2}{N-1} \rightarrow \frac{N-1}{N-2} \rightarrow \infty$
 $N \rightarrow \infty \Rightarrow N \approx N-1 \approx N-2$
 (Bollobás) numbers

increases
 $N \rightarrow \infty \Rightarrow N \approx N-1 \approx N-2$

$$\pi_{y,u}[2] = \frac{1}{N-2} \sum_{n=3}^N y[n]u[n-2]$$

$$\Rightarrow \sum_{n=3}^N y[n]u[n-2] = (N-2) \pi_{y,u}[2] \quad \left| \cdot \frac{1}{N-1} \right. = \frac{N-2}{N-1} \pi_{y,u}[2]$$

$$\hookrightarrow \approx \pi_{y,u}[2] \delta$$

$$\hat{\theta}_N = \begin{pmatrix} \pi_y^{N-1} \tau_0 & \pi_y^N \tau_1 & -\pi_{y_u}^{N-1} \tau_0 - \pi_{y_u}^N \tau_1 \\ \pi_y^N \tau_0 & \pi_y^{N-1} \tau_1 & -\pi_{y_u}^N \tau_1 - \pi_{y_u}^{N-1} \tau_0 \\ \hline -\pi_{y_u}^{N-1} \tau_0 - \pi_{y_u}^N \tau_1 & \pi_u^{N-1} \tau_0 & \pi_u^N \tau_1 \\ -\pi_{y_u}^N \tau_1 - \pi_{y_u}^{N-1} \tau_0 & \pi_u^N \tau_0 & \pi_u^{N-1} \tau_1 \end{pmatrix} \cdot \begin{pmatrix} \frac{N-2}{N} \\ \frac{N}{N-1} \end{pmatrix} \cdot \begin{pmatrix} -\pi_{y_u}^{N-1} \tau_1 \\ -\pi_{y_u}^N \tau_2 \\ \pi_{y_u}^N \tau_1 \\ \pi_{y_u}^{N-1} \tau_2 \end{pmatrix}$$

$$\chi_N^2 = \frac{1}{N} \sum_{u=1}^N (y_{\tau u} - \epsilon^T \tau u) \hat{\theta}_N)^2$$

Prüfung soll in der Vorlesung über, wenn etwas produziert $\epsilon \tau u \tau u$
 ist die Lösung in 0 (ergibt sich aus $\text{ARX}(2,2) \rightarrow$ ~~ergibt sich~~
~~ergibt sich~~ ~~ergibt sich~~

$$\text{Ic} \Rightarrow \sum \frac{1}{N} \epsilon \tau u \tau u \Rightarrow \epsilon \tau u \tau u = \begin{bmatrix} y_{\tau u-1} \\ y_{\tau u-2} \\ u_{\tau u-1} \\ u_{\tau u-2} \end{bmatrix} \epsilon \tau u$$

$$\begin{aligned} & \frac{1}{N} \sum y_{\tau u-1} \epsilon \tau u \\ & \frac{1}{N} \sum y_{\tau u-2} \epsilon \tau u \\ & \frac{1}{N} \sum u_{\tau u-1} \epsilon \tau u \\ & \frac{1}{N} \sum u_{\tau u-2} \epsilon \tau u \end{aligned} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{matrix} \text{ergibt sich} \\ \text{ergibt sich} \\ \text{ergibt sich} \\ \text{ergibt sich} \end{matrix} \begin{bmatrix} y_{\tau u-1} \\ y_{\tau u-2} \\ u_{\tau u-1} \\ u_{\tau u-2} \end{bmatrix} \epsilon \tau u$$

Exercițiu 4.3 - identificare pentru OE(1,1)

$$OE: A(z^{-1}) y[n] = B(z^{-1}) u[n] + \underbrace{A(z^{-1}) e[n]}_{\text{}}$$

$$y[n] = \frac{B(z^{-1})}{A(z^{-1})} u[n] + e[n]$$

$$\rightarrow (1 + a_1 z^{-1}) y[n] = b_1 z^{-1} u[n] + (1 + a_1 z^{-1}) e[n]$$

$$y[n] + a_1 z^{-1} y[n] = b_1 u[n-1] + a_1 e[n-1] + e[n]$$

$$y[n] = a_1 (e[n-1] - y[n-1]) + b_1 u[n-1] + e[n]$$

$$y[n] = e^T \theta + e[n] \quad (\text{formă de regresie liniară})$$

Se va deservi cum zgomotul afectează direct datele de test

$$\begin{cases} e^T = [e[n-1] - y[n-1]; u[n-1]] \\ \theta^T = [a; b] \end{cases}$$

✓ MCMMP - se poate aplica
dacă se poate modela
separat zgomotul alb + dacă
se poate estima

Estimarea CMMP:

$$\hat{\theta}_N = \left[\frac{1}{N} \sum_{n=1}^N e[n] e^T[n] \right]^{-1} \left[\frac{1}{N} \sum_{n=1}^N e[n] y[n] \right]$$

↳ Vom considera zgomotul
estimat $\{e[n]\}_{n=1, \dots, N}$

$$\begin{aligned} \circ \text{Decoree } R_{yy}[k] &= E \{ e[n] y[n-k] \} = E \left\{ e[n] \cdot \left(\frac{B(z^{-1})}{A(z^{-1})} u[n-k] + e[n-k] \right) \right\} \\ &= \frac{B(z^{-1})}{A(z^{-1})} R_{eu}[k] + \lambda^2 \delta_0[k], \quad k \in \mathbb{Z} \end{aligned}$$

descrie este practic necorelată cu zgomotul alb dacă informația este necorelată
cu zgomotul alb

$$\frac{N}{N-1} R_N = \begin{bmatrix} \pi_{ey}^{N-1}[\omega] & + \pi_{eu}^{N-1}[\omega] - \pi_{yu}^{N-1}[\omega] \\ + \pi_{eu}^{N-1}[\omega] - \pi_{yu}^{N-1}[\omega] & \pi_{uu}^{N-1}[\omega] \end{bmatrix}; \quad \frac{N}{N-1} \mu_N = \begin{bmatrix} \pi_{ey}^{N-1}[\omega] - \pi_{yu}^{N-1}[\omega] \\ \pi_{yu}^{N-1}[\omega] \end{bmatrix}$$

$$\Delta_N = \pi_{uu}^{N-1}[\omega] \pi_{uu}^{N-1}[\omega] - (\pi_{eu}^{N-1}[\omega] - \pi_{yu}^{N-1}[\omega])^2$$

$$\hat{a}_N = \frac{\pi_{eu}^{N-1}[\omega] (\pi_{ye}^N[\omega] - \pi_{ye}^N[\omega]) + \pi_{yu}^N[\omega] (\pi_{yu}^{N-1}[\omega] - \pi_{eu}^{N-1}[\omega])}{\Delta_N}$$

$$\hat{b}_N = \pi_{ey}^{N-1}[\omega] \pi_{yu}^N[\omega] - \pi_{yu}^N[\omega] \pi_{yu}^{N-1}[\omega] + \pi_{yu}^{N-1}[\omega] \pi_{ye}^N[\omega] - \pi_{eu}^{N-1}[\omega] \pi_{ye}^N[\omega] + \pi_{eu}^{N-1}[\omega] \pi_{yu}^N[\omega]$$

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Exaktion auslösen

$$\pi_{ye}[\omega] = \frac{B(z^{-1})}{A(z^{-1})} \pi_{ue}[\omega] + \chi^2 \delta_0[\omega]$$

$$\hookrightarrow \pi_{ye}[\omega] = \frac{B(z^{-1})}{A(z^{-1})} \pi_{ue}[\omega]$$

$$\pi_{e-y}[\omega] \in \{(\pi_{eu}[\omega] - \pi_{yu}[\omega])^2\} = \chi^2 \delta_0[\omega] - 2 \pi_{ye}[\omega] + \pi_{yu}[\omega]$$

$$= \pi_{yu}[\omega] - \chi^2 - 2 \frac{B(z^{-1})}{A(z^{-1})} \pi_{ue}[\omega]$$

$$\pi_{yu}[\omega] = \{ \pi_{yu}[\omega] \pi_{yu}[\omega] \} = \{ \frac{B(z^{-1})}{A(z^{-1})} \pi_{ue}[\omega] + \pi_{ue}[\omega] \}$$

$$= \{ \frac{B(z^{-1})}{A(z^{-1})} \pi_{ue}[\omega] + \pi_{ue}[\omega] \} \left(\frac{B(z^{-1})}{A(z^{-1})} \pi_{ue}[\omega] + \pi_{ue}[\omega] \right)$$

$$\pi_{yu}[\omega] = \pi_{yu}[\omega] = \frac{B(z^{-1})}{A(z^{-1})} \pi_{ue}[\omega] + \pi_{ue}[\omega]$$

• Zimultane Gleichungen:

$$\theta = \begin{bmatrix} a \\ b \end{bmatrix} = [E \{ e(n) e^T(n) \}]^{-1} [E \{ e(n) y(n) \}] - [E \{ e(n) e^T(n) \}]^{-1} [E \{ e(n) u(n) \}]$$

$$E \{ e(n) e^T(n) \} = \begin{bmatrix} \sigma_e^2 & \sigma_{eu}(z) - \sigma_{yu}(z) \\ \sigma_{eu}(z) - \sigma_{yu}(z) & \sigma_u^2 \end{bmatrix}$$

$$E \{ e(n) y(n) \} = \begin{bmatrix} \sigma_{ye}(z) - \sigma_{ye}(z) \\ \sigma_{yu}(z) \end{bmatrix}, \quad E \{ e(n) u(n) \} = \begin{bmatrix} -\sigma_{ye}(z) \\ \sigma_{yu}(z) \end{bmatrix}$$

• Condition de convergence (c'est $\theta_n = \theta$):

$$\begin{cases} a) E \{ e(n) e^T(n) \} > 0 \text{ (invertible)} \\ b) \sigma_{ye}(z) = 0, \sigma_{yu}(z) = 0 \\ c) u = \text{signal blanc } (0, \sigma^2) \end{cases} \quad z^{-2} \sigma_{ye}(z) = \sigma_{ye}(z)$$

$$\sigma_{ye}(z) = 0 \Leftrightarrow \frac{B(z^{-1})}{A(z^{-1})} \sigma_{yu}(z) = 0 \Leftrightarrow b_2^{-1} (1 + d_1 z^{-1} + d_2 z^{-2} + \dots) \sigma_{yu}(z) = 0$$

$$\Rightarrow b \sigma_{yu}(z) + b d_1 \sigma_{yu}(z^{-1}) + b d_2 \sigma_{yu}(z^{-2}) + \dots = 0$$

• Condition suffisante naturelle: $u = \text{signal blanc } (0, \sigma_u^2)$, nous obtiendrons

$$\sigma_{ye}(k) = \lambda^2 \sigma_0(k)$$

$$\sigma_{ye}(z) = \sigma_{yu}(z) - \lambda^2$$

$$\sigma_{yu}(k) = \frac{B(z^{-1})}{A(z^{-1})} \sigma_u^2 \sigma_0(k) =$$

$$\sigma_{ye}(z) = 0$$

$$\sigma_{yu} = 1 = b \sigma_u^2$$

$$x_y[k] + a x_y[k-1] = b x_y[k-1] + x_y[k] + a x_y[k-1]$$

$$\underline{k=0}: x_y[0] + a x_y[-1] = b x_y[-1] + x_y[0] + a x_y[-1]$$

$$= b x_y[-1] - b^2 v_n^2 \frac{\lambda^2}{1+\lambda^2}$$

$$\underline{k=1}: x_y[1] + x_y[0] = b x_y[0] + x_y[1] + a x_y[0]$$

$$\Rightarrow a x^2$$

$$(1-a^2) x_y[0] - b^2 v_n^2 + \lambda^2 - a^2 \lambda^2 \Rightarrow x_y[0] = x^2 + \frac{b^2}{1-a^2} v_n^2$$

$$x_y[1] = a(\lambda^2 - x_y[0]) = \frac{a b^2}{a^2 - 1} v_n^2$$

$$\begin{aligned} \begin{bmatrix} x_y[0] & x_y[1] \end{bmatrix} &= \begin{bmatrix} \frac{b^2}{1-a^2} v_n^2 & 0 \\ 0 & v_n^2 \end{bmatrix} \\ \begin{bmatrix} x_y[0] & y_y[0] \end{bmatrix} &= \begin{bmatrix} -\frac{a b^2}{a^2 - 1} v_n^2 \\ b v_n^2 \end{bmatrix} \end{aligned} \Rightarrow \begin{bmatrix} 94 \\ 6 \end{bmatrix} = \begin{bmatrix} x_y[0] & x_y[1] \\ x_y[0] & y_y[0] \end{bmatrix} \times \begin{bmatrix} 94 \\ 6 \end{bmatrix}$$

OE [2:2]:

$$\left. \begin{aligned} A(z^{-1}) y(n) &= B(z^{-1}) u(n) + A(z^{-1}) e(n) \\ A(z^{-1}) &= 1 + a_1 z^{-1} + a_2 z^{-2} \\ B(z^{-1}) &= b_1 z^{-1} + b_2 z^{-2} \end{aligned} \right\} \Rightarrow$$

$$\Rightarrow y(n) + a_1 y(n-1) + a_2 y(n-2) = b_1 u(n-1) + b_2 u(n-2) + e(n) + a_1 e(n-1) + a_2 e(n-2)$$

$$\Rightarrow y(n) = a_1 (e(n-1) - y(n-1)) + a_2 (y(n-2) - y(n-1) + b_1 u(n-1) + b_2 u(n-2))$$

$$\hookrightarrow y(n) = \underbrace{[e(n-1) - y(n-1) \quad e(n-2) - y(n-1) \quad e(n-1) \quad u(n-1)]}_{e^T} \begin{bmatrix} a_1 \\ a_2 \\ b_1 \\ b_2 \end{bmatrix} + e(n)$$

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$y(n) = e^T \theta + e(n) \Rightarrow$ forma di regressione lineare

$$\Rightarrow \left\{ \begin{aligned} \hat{\theta} &= \left[\frac{1}{N} \sum_{n=1}^N e(n) e^T(n) \right]^{-1} \left[\frac{1}{N} \sum_{n=1}^N e(n) y(n) \right] \\ \hat{\theta} &= R_N^{-1} p_N \end{aligned} \right.$$

$$\begin{aligned} \text{cov}[k] &= E\{e(n) y(n-k)\} = E\left\{e(n) \left(\frac{B(z^{-1})}{A(z^{-1})} u(n-k) + e(n-k) \right)\right\} \\ &= \frac{B(z^{-1})}{A(z^{-1})} \pi_{ku}[k] + \sigma_e^2 \delta[k] \end{aligned}$$

$$\frac{N}{N-1} R_N = \begin{pmatrix} \pi_{e-y}[0] & \pi_{e-y}[1] & \pi_{eu}[0] - \pi_{ey}[0] & \pi_{eu}[1] - \pi_{ey}[1] \\ \pi_{e-y}[1] & \pi_{e-y}[2] & \pi_{eu}[1] - \pi_{ey}[1] & \pi_{eu}[2] - \pi_{ey}[2] \\ \pi_{eu}[0] - \pi_{ey}[0] & \pi_{eu}[1] - \pi_{ey}[1] & \pi_{uu}[0] & \pi_{uu}[1] \\ \pi_{eu}[1] - \pi_{ey}[1] & \pi_{eu}[2] - \pi_{ey}[2] & \pi_{uu}[1] & \pi_{uu}[2] \end{pmatrix}$$

es: $\pi_{e-y}[0] = \frac{1}{N-1} \sum_{u=1}^{N-1} (\pi_{eu}[0] - \pi_{ey}[0])^2 = \frac{1}{N-1} \sum_{u=2}^N (\pi_{eu}[0] - \pi_{ey}[0])^2$

$$\pi_{e-y}[1] = \frac{1}{N-1} \sum_{u=2}^N (\pi_{eu}[1] - \pi_{ey}[1]) (\pi_{eu}[0] - \pi_{ey}[0])$$

$$\pi_{eu}[0] - \pi_{ey}[0] = \frac{1}{N-1} \left(\sum_{u=2}^N \pi_{eu}[0] + \pi_{eu}[1] \right) - \sum_{u=2}^N \pi_{ey}[0] + \pi_{ey}[1]$$

$$\frac{N}{N-1} R_N = \begin{bmatrix} \pi_{ye}[1] - \pi_{yy}[1] \\ \pi_{ye}[2] - \pi_{yy}[2] \\ \pi_{yeu}[1] \\ \pi_{yeu}[2] \end{bmatrix}$$

$$\rightarrow \theta_N = R_N^{-1} \cdot \pi_N$$

Dann $u \sim \text{i.i.d. } (0, \sigma_u^2)$

$\Rightarrow$ conditione de la base indépendante

Conditione de covariance

a) $E\{\pi_{eu} \pi_{ey}\} \Rightarrow$ universelles

$$\chi_N^2 = \frac{1}{N} \sum_{u=1}^N (\pi_{eu} - \pi_{eu} \theta_N)^2$$

b) $E\{\pi_{eu} \pi_{eu-1}\} = \pi_{eu}[1] = 0$

(bede notwendig an differenz)
ignorant

c) $e \sim \text{ignorant}$

$$E\{\pi_{eu} \pi_{eu-1}\} = \pi_{eu}[1] = 0$$

$$\pi_{ey}[2] = 0$$

also $(0, \sigma^2)$

$$E\{\pi_{eu} \pi_{eu-1}\} = \pi_{ey}[2] = 0$$

more dispense